AASRA: Agronomic ML Architecture & Vertex AI Guide

Specialised ML Models for PS-02 + PS-03 + PS-04 + PS-07 & Google Cloud Deployment

Scope Alignment: Hack Core 2026 Problem Statements (PS-02, PS-03, PS-04, PS-07) | **Cloud Stack:** Google Vertex AI / Cloud Run + Next.js | **Version:** 1.0 Production Blueprint

1. System Architecture & The Modular ML Principle

A foundational principle defined by the hackathon organizers is: **'The selected solution should use specialised ML models for risk, recommendation, yield and causal attribution, while Gemini acts as the multilingual conversational layer. We should NOT train one giant model.'**

Training a single monolithic model fails in agriculture because physical biophysics, chemical label safety, and causal economics require different learning paradigms. AASRA implements a modular, decoupled architecture where each model has an explicit, verifiable role and feeds structured outputs into the Gemini conversational layer and website UI.

Problem Statement	Model / Component	Core Algorithm / Type	Target Output & Purpose
PS-02	**Climate Stress Prediction**	XGBoost / LightGBM Classifier	Predicts upcoming heat, drought, heavy rain probability (0-100%), time window (days) and severity.
PS-02	**Biological Intervention Readiness**	Hybrid Rule Engine + Scorer	Converts risk to action: Readiness Score (0-100), APPLY NOW / WAIT / AVOID status, 36h spray window.
PS-03	**Biological Product Ranking**	Hard Filter + LGBM Ranker	Ranks 50 Syngenta biologicals for crop, stage, soil, and stress compatibility with fit score.
PS-03 / PS-07	**Expected Product Response**	Treatment-Response Regressor	Estimates stress reduction range (23-35%) and expected yield impact (+5-9%) with uncertainty.
PS-07	**Field Yield Prediction**	Gradient-Boosted Regressor	Predicts expected field-level yield (e.g. 20.4 q/acre; range 18.7-22.1) and confidence.
PS-07	**Causal Biological Impact / ROI**	Causal ML Meta-Learner (T-Learner)	Counterfactual lift: true incremental yield directly attributable to biological (+2.2 q/acre) & ROBI.
PS-04	**Gemini Multilingual Layer**	Gemini 2.5 Flash + Tool Calling	Understands farmer voice/text in local languages, orchestrates specialised models, explains 'Why?'.

Architectural Mandate: Gemini acts as the intelligent orchestration, reasoning, and explanation bridge. It translates farmer queries, triggers the specialised ML models via structured JSON parameters, and narrates model outputs with transparent 'Why?' explanations.

2. PS-02: Climate Stress Prediction & Intervention Timing

Model 1: Climate Stress Prediction Model (PS-02)

Purpose: Predict the probability, timing, and severity of climate-related crop stress for a specific field.
Inputs: Location (lat/lon, district), crop, crop stage, sowing date, daily TMax, TMin, humidity, rainfall, wind, solar radiation, volumetric soil moisture (0-28cm), soil texture/pH, historical weather anomalies, and season/month.
Outputs: Stress type, probability (0-100%), arrival window (days), severity level, confidence score.
Example Output: *Heat Stress -> 78% probability -> Expected in 6-9 days -> Moderate severity.*

- **Recommended Algorithm:** LightGBM or XGBoost Classifier. We construct rolling feature windows over 3, 7, and 14 days: consecutive dry spell days, Heat Stress Index (HSI), and Vapor Pressure Deficit (VPD).
  - **Training Target:** Ground-truth stress events or defensible biophysical proxy labels: Thermal spike ($T_{\text{night}} \ge 24.5^\circ\text{C}$ or $T_{\max} \ge 35^\circ\text{C}$), Drought ($ ext{VPD} \ge 2.8\text{ kPa}$ & Soil Moisture $<25\%$), and Waterlogging ($>50\text{ mm}$ rainfall on heavy clay).

Model 2: Biological Intervention Timing / Readiness Model (PS-02)

Purpose: Convert climate risk predictions and environmental conditions into an actionable biological intervention window.
Core Philosophy: A biological intervention must be applied *proactively before* the crop reaches irreversible cellular damage, but strictly under safe application conditions (avoiding wind drift or rain washoff).

Condition Evaluated	Threshold / Rule	Readiness Impact
Spray Rain Feasibility	Precipitation probability < 20% in next 6h	Rain >5mm washes foliar actives off leaf surface (AVOID/WAIT).
Wind Drift Limit	Wind speed < 15 km/h	Wind >15 km/h causes droplet drift and uneven canopy coverage.
Heat Evaporation Cap	Temperature at spray time < 35°C	High ambient heat evaporates droplets before stomatal absorption.
Phenological Vulnerability	Crop at flowering / pod-set / squaring	Reproductive window provides highest biostimulant response.
Optimal Diurnal Window	6:00 AM - 9:30 AM or 4:30 PM - 6:30 PM	Stomata are open and transpiration pull is optimal.

Model Architecture: Hybrid Model = Deterministic Agronomic Guardrails (Hard Constraints) + Trained Scoring Regressor (LightGBM) outputting `readiness_score` (0-100).
Output Example: *Readiness Score 84/100 -> Status: APPLY NOW -> Optimal Window: Next 36 Hours (6:00 AM - 9:30 AM).*

Architectural Mandate: By pairing Climate Stress Prediction with Intervention Readiness, AASRA never gives a generic alert. It tells the farmer exactly: What stress is coming, whether today is safe to spray, and the exact 36-hour operational window.

3. PS-03 & PS-07: Product Ranking, Yield & Causal Impact

Model 3: Biological Product Recommendation & Ranking (PS-03)

Purpose: Rank the best biological product for a specific farmer, crop, soil, growth stage, and observed stress context.
Two-Stage Architecture:
1. **Hard Disqualifier Filters:** Disqualifies products by wrong crop label, incompatible phenology stage, invalid application method, or known chemical antagonism (e.g. never mix biologicals with alkaline copper fungicides).
2. **ML Ranking Engine (LGBMRanker):** Scores candidate products based on: stress suitability, crop compatibility, growth stage match, soil moisture/pH, historical trial efficacy, and expected net economic return.

Model 4: Expected Product Response Model (PS-03 / PS-07)

Mandate: *Show ranges and uncertainty. Do not present an expected response as a guaranteed effect.*
Inputs: Field conditions, crop, stage, stress severity, weather, product dose, and application method.
Outputs: Expected stress reduction range (e.g. *23-35%*), expected yield impact range (e.g. *+5-9%*), and Confidence: *Medium (84%)*.

Model 5: Field-Level Yield Prediction Model (PS-07)

Purpose: Predict expected yield for a specific field accounting for crop variety, sowing date, weather trajectory, soil characteristics, satellite vegetation indices (NDVI/NDWI), and farmer interventions.
Model: Gradient-boosted regression (XGBoost) with *field/season-aware cross-validation* to prevent spatial data leakage.
Output Example: *Expected Yield: 20.4 q/acre; Range: 18.7 - 22.1 q/acre; Confidence: 82%.*

Model 6: Causal Biological Impact / ROI Model (PS-07)

The Core Challenge: Before/after or simple treated-vs-untreated averages are NOT causal attribution. Treated fields often receive better water or fertilizer, creating severe confounding bias. The system must answer: *What would the yield have been WITHOUT the biological versus WITH the biological under identical conditions?*

Causal ML Formulation: T-Learner (Two-Learner Meta-Model)

```
# Train Base Learner 0 on Control (Untreated) Fields: mu0(X) = E[Y | X, Treatment=0]
# Train Base Learner 1 on Treated Fields: mu1(X) = E[Y | X, Treatment=1]
counterfactual_yield = model_mu0.predict(field_features) # e.g. 18.2 q/acre
factual_yield = model_mu1.predict(field_features) # e.g. 20.4 q/acre
causal_treatment_effect (CATE) = factual_yield - counterfactual_yield # +2.2 q/acre (+12.1%)
net_causal_profit = (2.2 * mandi_price_per_qtl) - biological_product_cost
```

Architectural Mandate: The T-Learner isolates the true incremental lift produced by the biological product, mathematically proving to the farmer and judges that the yield gain was caused by the intervention, not random weather luck.

4. PS-04: Gemini Conversational Layer & Training Pipeline

Model 7: Gemini Conversational Layer (PS-04)

Role: The user-facing multilingual voice and chat interface. We use **Gemini 2.5 Flash** with structured tool/function calling rather than training a generic LLM from scratch.

Responsibilities:

1. **Multilingual Understanding:** Handles Hindi, Marathi, Punjabi, Telugu, English, and regional dialects.
2. **Intent & Entity Extraction:** Extracts field location, crop type, sowing date, and observed symptoms.
3. **Model Orchestration:** Calls PS-02 Stress Model, PS-03 Recommendation Model, and PS-07 Causal Model.
4. **Empirical Explanation:** Explains model outputs in simple, actionable farmer language (answering *'Why am I getting this recommendation?'*).

Core Training Record Specification

According to the hackathon dataset standard, every observation in our training matrix follows this unified schema:

Category	Features Ingested	Data Source
Field Profile	Field ID, lat/lon, district, area (acres), crop, variety, sowing date, irrigation status	AASRA Field Store / Polygon GPS
Soil Properties	Texture (sand/clay %), moisture (0-28cm), pH, EC (salinity), Organic Carbon, N, P, K	ISRIC SoilGrids 2.0 & NASA SMAP
Weather Telemetry	TMax, TMin, humidity, rainfall, wind, solar radiation, VPD, GDD, 16-day forecast	Open-Meteo & ERA5-Land (ECMWF)
Crop State	Growth stage (vegetative/bloom/pod), DAS, NDVI, NDWI, historical stress days	Sentinel-2 L2A & FAO-56 Phenology
Intervention	Product applied (Quantis/Ampligo), application date, dose/acre, method, area treated	AASRA Field Action Journal
Outcome Labels	Observed stress (type/severity), response confirmation, final harvest yield (q/acre)	ICAR AICRP Trials & Field Verifications

Python Training Script Architecture (`scripts/train_hackathon_models.py`)

Python Model Pipeline (LightGBM, XGBoost, Causal T-Learner)

```
from lightgbm import LGBMClassifier, LGBMRegressor
from xgboost import XGBRegressor
# 1. PS-02: Climate Stress Classifier
stress_model = LGBMClassifier(n_estimators=150, learning_rate=0.05, max_depth=6)
stress_model.fit(X_train, y_stress_labels)
# 2. PS-07: Causal T-Learner for Biological Lift
X_control = X[treatment == 0]; y_control = y[treatment == 0]
X_treated = X[treatment == 1]; y_treated = y[treatment == 1]
mu0_model = XGBRegressor().fit(X_control, y_control) # Yield under No Product
mu1_model = XGBRegressor().fit(X_treated, y_treated) # Yield under Biological Product
joblib.dump({'stress': stress_model, 'mu0': mu0_model, 'mu1': mu1_model}, 'models/aasra_ml_suite.joblib')
```

5. Google Cloud Deployment & Next.js Website Integration

Cloud Architecture Comparison: Vertex AI vs Google Cloud Run

Google Cloud Platform provides two methods to serve these models online with a public HTTPS endpoint:

Feature / Metric	Google Vertex AI Endpoint	Google Cloud Run (Serverless Container)
Deployment Model	Managed Vertex AI Model Registry + Node Endpoint	Serverless FastAPI Docker Container on GCP
Billing Structure	24/7 VM reservation (minimum ~$50-70/month)	**Scales to 0 when idle ($0 cost when not querying!)**
Latency / Response	15-30ms dedicated inference	25-45ms inference (sub-50ms)
API Protocols	Google gRPC & Vertex REST (requires IAM token)	**Standard HTTPS JSON REST (native for Next.js fetch)**
Recommendation	Best for enterprise 24/7 high-throughput fleets	**RECOMMENDED for Hackathon & Web Launch**

Step-by-Step Deployment Commands (GCP)

Deploying the Model Suite to Google Cloud (One-Command Deployment)

```
# Step 1: Package FastAPI app with trained .joblib models into Docker container
# Step 2: Deploy container directly to Google Cloud Run (Asia South - Mumbai):
gcloud run deploy aasra-ml-suite \
 --source . \
 --region asia-south1 \
 --memory 2Gi \
 --cpu 2 \
 --min-instances 0 \
 --allow-unauthenticated
# Result: Live HTTPS Endpoint -> https://aasra-ml-suite-xxxxxx.a.run.app/predict
```

Live Website Integration (Next.js Frontend)

The live Next.js website connects to the Google Cloud ML service via a secure server route (`frontend/src/app/api/ml/predict/route.ts`). If Google Cloud is cold-starting or unreachable, the system automatically falls back to our calibrated biophysical engine, guaranteeing **100% uptime with zero crashes** for farmers and evaluators.

Hackathon Execution Priority Roadmap

Priority	Milestone & Components	Deliverables
P0 — Core	PS-02 Climate Stress + PS-03 Product Ranking + PS-04 Gemini + PS-07 Causal ROI	Trained LightGBM & T-Learner models, live Next.js API integration.
P1 — Strong	Yield Prediction + Biological Readiness Scorer + Syngenta Product KB	36h spray window calculation, field-aware cross-validation.
P2 — Extension	Crop-Photo Intelligence (Gemini Vision) + Advanced Response Calibration	Visual foliar verification and multi-year economic balance sheet.

Architectural Mandate: The complete live web application is accessible at: https://frontend-phi-flame-21.vercel.app. This document serves as the official technical architecture guide for all Hack Core 2026 problem statements.